

Open Source Reproduction of Scaling Laws, Tabular Data and Actuarial Ratemaking Models

Ronald Richman

May 3, 2026

1 FMTPL Reproduction Design and Results

This supplementary material describes a smaller-scale reproduction of the empirical protocol used in the paper. The main experiments in the paper use a proprietary multi-million-row motor portfolio. To make the methodology inspectable and runnable by practitioners, we use the French MTPL (FMTPL) frequency data, where MTPL denotes motor third-party liability. The data are from `CASdatasets` by Dutang and Charpentier (2026); we use the corrected version documented by Wüthrich and Merz (2023). This dataset has been extensively studied in the recent actuarial pricing and actuarial machine-learning literature, making it a useful benchmark for a reproducible illustration. The experiment is intended to reproduce the *experimental design* and modeling ideas of the paper, not the numerical results obtained on the proprietary portfolio. The accompanying repository, including the Colab notebook, source code, data-preparation script, reference outputs, and figures, is available at <https://github.com/RonRichman/frmtpl-scaling-laws>.

The remainder of this supplementary material is organized as follows. Section 1 describes the FMTPL data source, the reduced experimental design, the compact model families, the reference scaling results, and the business-facing outcome diagnostics. Section 2 gives a short tutorial-style guide to the GitHub repository and the commands needed to rerun the workflow.

1.1 Purpose and Relation to the Main Experiment

The reproduction keeps the main actuarial and scaling-law structure unchanged. Each policy-period record has an exposure E_i and claim count y_i , and every model predicts an exposure-scaled Poisson mean

$$\hat{\lambda}_i = E_i \exp(f_\theta(x_i)). \quad (1)$$

Performance is evaluated by held-out mean Poisson deviance, and data scaling is studied by fitting

$$L(N) \approx L_\infty + A_N N^{-\alpha}, \quad (2)$$

where N is the number of training rows in a nested subsample. As in the main paper, α is interpreted as a data-scaling exponent: larger values indicate faster improvement toward the fitted floor within the observed regime.

The experiment therefore gives actuaries a compact way to inspect the mechanics of the study: how exposure enters the loss, how nested subsamples are formed, how random seeds are averaged,

and how scaling exponents are estimated from the resulting performance summaries. Because the FMTPL data are much smaller and less feature-rich than the proprietary data, the resulting model rankings and fitted exponents should not be read as estimates of the production-scale portfolio studied in the paper.

1.2 Data Source, Cleaning and Split

The FMTPL data contain motor third-party liability policy records from a French portfolio. Within `CASdatasets`, the technical dataset name is `freMTPL2`. The frequency file `freMTPL2freq` contains policy and claim-frequency information, while `freMTPL2sev` contains the corresponding claim-severity information.¹ For the reproduction, we use the corrected `freMTPL2freq.rda` frequency file available at <https://people.math.ethz.ch/~wueth/Lecture/freMTPL2freq.rda>. This version follows the cleaning described in Appendix B of Wüthrich and Merz (2023), including the corrected claim counts obtained from the severity file, the removal of policies with more than five claims, exposure censoring at one year, and the retained pre-2016 region labels.² Variable definitions and feature descriptions are documented in `CASdatasets`; the correction and cleaning choices used here are documented in Appendix B of Wüthrich and Merz (2023). The experiment uses the learning/test split in Listing 5.2 of Wüthrich and Merz (2023). It uses `RNGversion("3.5.0")` and `set.seed(500)`, producing $|\mathcal{L}| = 610,206$ learning records and $|\mathcal{T}| = 67,801$ test records. The extra `sample_unif` column is not part of the source data; it is added only to create nested learning subsets for the scaling-law experiment while preserving the fixed test set. The claim amount field is excluded, so the experiment studies frequency only. When using this reproduction, cite the `CASdatasets` source and the correction documented by Wüthrich and Merz (2023).

1.3 Reduced Experimental Design

The reproduction uses the same conceptual protocol as the main paper but with smaller models and fewer seeds. The default experiment trains at fractions

$$f \in \{0.05, 0.10, 0.25, 0.50, 0.75, 1.00\}$$

of the training split, using a fixed `sample_unif` variate so that smaller datasets are nested subsets of larger datasets. For each model and fraction, three independent seeds are trained and their predictions are averaged before evaluating the ensemble deviance. This mirrors the seed-averaged ensemble score used in the main experiment, while keeping the run computationally modest.

1.4 Compact Model Families

The reproduction includes five model families that summarize the main themes of the paper:

¹Wüthrich and Merz (2023) note that the FMTPL data are distributed through the R package `CASdatasets`, hosted by Dutang and Charpentier, with documentation at <http://cas.uqam.ca/>.

²Wüthrich and Merz (2023) use `CASdatasets` version 1.0-8, packaged on 2018-05-20. They prefer this older version because it retains the 22 French region labels R11, ..., R94. Later `CASdatasets` versions replace these labels with region names and merge R31 (Nord-Pas-de-Calais) and R41 (Lorraine) into a single Nord-Pas-de-Calais region, which they regard as an error. They note that the older version can be obtained in R using `library(OpenML)`, `library(farff)`, and `freMTPL2freq <- getOMLDataSet(data.id = 41214)$data`.

Design element	Main paper	FMTPL reproduction
Data source	Proprietary motor insurance portfolio	FMTPL frequency data
Target	Claim frequency	Claim frequency
Exposure handling	Multiplicative Poisson offset E_i	Multiplicative Poisson offset E_i
Training fractions	Nested fractions from 5% to 100%	Same default fraction grid
Seeds	Five seeds per configuration	Three seeds per configuration
Model capacity	Research-scale model ladders, including large TabM and Transformer variants	Compact Keras model configurations
Result interpretation	Production-scale scaling evidence for the studied portfolio	Transparent reproduction of methodology and model family behavior

Table 1: Comparison of the proprietary experiment and the FMTPL reproduction.

- **GLM**: a Keras GLM-style log-link baseline using one-dimensional embeddings and an exposure-scaled exponential output head.
- **FFN**: a deliberately small, one-hidden-layer feed-forward neural network over flattened entity embeddings, representing the standard tabular MLP baseline while remaining stable on the smallest FMTPL training cuts.
- **MultiCLS Transformer**: a compact tabular Transformer with multiple learned CLS pooling tokens, included to demonstrate attention-based interaction modeling and the CLS bottleneck idea.
- **MultiCLS+SSL Transformer**: the same broad Transformer recipe augmented with row-swap corruption detection, giving a compact version of the paper’s self-supervised representation-learning theme.
- **TabM-mini**: a parameter-efficient within-model ensemble with lightweight multiplicative adapters and a shared MLP backbone, included because TabM-style models are central to the main paper’s strongest scaling results (Gorishniy, Kotelnikov, & Babenko, 2024).

All neural models use the same preprocessed feature representation. Categorical variables are mapped to integer levels, continuous variables are discretized into quantile bins, and each feature is represented by a learned embedding. The quantile binning is deliberate. It aligns the reproduction with the main paper’s modeling representation, where the proprietary portfolio did not expose the relevant numerical covariates as raw continuous inputs to the model pipeline, but rather as quantile-binned covariates. This choice also makes the neural-network models less performant than they would be with richer continuous-variable inputs, which helps explain why the metrics in this supplementary material are weaker than results reported in other publications on the FMTPL data. For this experiment, the bin cut-points are estimated on the learning split and then applied unchanged to the test split. This keeps the reproduction close to the information set used in the main paper and avoids giving the reproduction a richer continuous-covariate representation than the proprietary experiment used. More generally, this mirrors the

main paper’s choice to keep feature tokenization fixed across model families, so that differences in scaling behavior are attributable primarily to the model class and objective rather than to incompatible preprocessing pipelines.

1.5 Scaling-Law Quantities

The analysis records seed-level scores, seed-averaged ensemble scores, and log-linear fits of $L(N) - L_\infty$ against N for each model configuration. The fitted floor L_∞ is empirical and should be interpreted only within this experiment.

The figures emphasize interpretability for actuarial users. The data-scaling figure plots held-out Poisson deviance against training rows on a log scale. The GLM-lift figure converts deviance improvements to per-policy log-likelihood lift by dividing the deviance difference by two, following the likelihood interpretation used in the main paper.

1.6 Reference Results from the FMTPL Run

Table 2 reports the held-out mean Poisson deviance from the FMTPL reference run. Each entry is the score of the seed-averaged ensemble over three independent fits. The table should be read as a reference run for the reproduction and as an illustration of the scaling workflow; it is not intended to reproduce the proprietary-portfolio rankings reported in the main text.

Training fraction	GLM	FFN	MultiCLS	MultiCLS+SSL	TabM-mini
0.05	0.244639	0.245887	0.244629	0.243944	0.244173
0.10	0.243088	0.242307	0.243719	0.243322	0.242124
0.25	0.241807	0.240978	0.240755	0.241415	0.240659
0.50	0.241709	0.240424	0.240649	0.240729	0.240581
0.75	0.241626	0.240201	0.240223	0.240297	0.240001
1.00	0.241440	0.239948	0.240024	0.240128	0.239787

Table 2: Held-out mean Poisson deviance for the FMTPL reference run. Lower values are better. Each cell scores the average prediction over three seeds.

Figure 1 plots the same results against the number of learning rows. The results show that each family benefits from additional data, while the small-data ordering remains unstable and the FMTPL data set is too small to support the stronger production-scale claims made in the main paper.

Figure 2 presents the same run on the reducible-loss scale used for the power-law fits. For each model, the plotted quantity is $L(N) - L_\infty$, where L_∞ is the fitted floor for that model in Table 3. The fitted lines are therefore model-specific summaries of the scaling run, rather than a pooled law across model families.

Table 3 reports the fitted scaling-law summaries obtained from the six training fractions. The fitted floor L_∞ and exponent α are estimated from the empirical reference run and should be interpreted as descriptive summaries over this narrow range of training sizes, rather than as stable population-level constants.

The FMTPL exponents are not directly comparable to the main-paper exponents. The FMTPL experiment has fewer rows, fewer features, compact model configurations, and model-specific

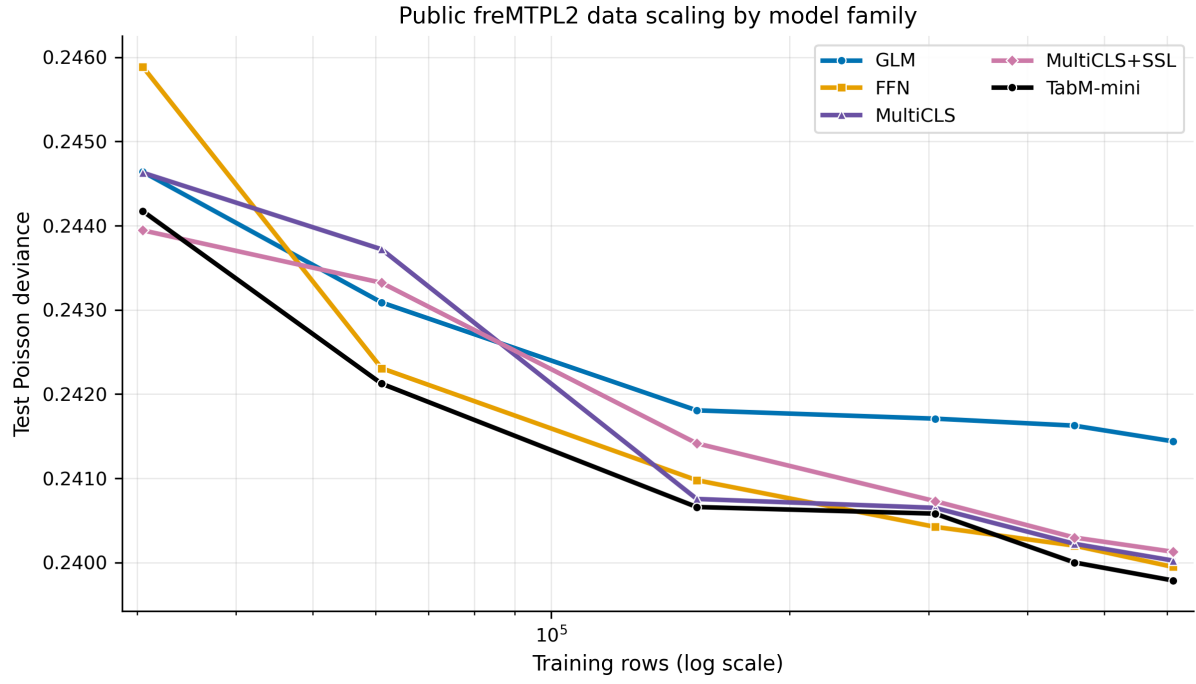


Figure 1: Data-scaling curves for the FMTPL reference run. Points show seed-averaged ensemble test Poisson deviance at each nested training fraction.

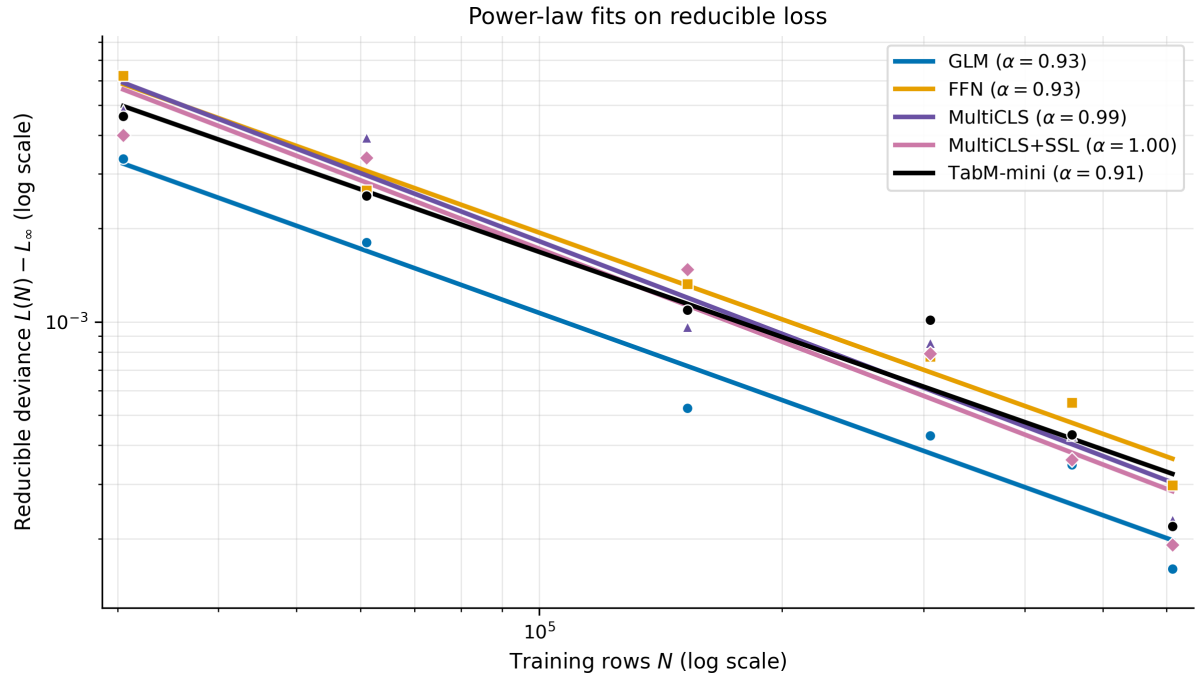


Figure 2: Power-law fits on the reducible-loss scale for the FMTPL reference run. Points show $L(N) - L_\infty$; lines show $A_N N^{-\alpha}$.

Model	L_∞	α	R^2
GLM	0.241280	0.933	0.961
FFN	0.239651	0.928	0.984
MultiCLS Transformer	0.239793	0.991	0.949
MultiCLS+SSL Transformer	0.239937	0.996	0.933
TabM-mini	0.239568	0.912	0.932

Table 3: Descriptive power-law fits $L(N) \approx L_\infty + A_N N^{-\alpha}$ for the FMTPL reference run.

fitted floors. The run is intended to reproduce the workflow and diagnostics, not to estimate the production-scale scaling constants.

Figure 3 reproduces the paper’s parameter-performance view using the compact FMTPL model set. The experiment has one compact configuration per family rather than a full size ladder, so the figure should be read as a full-data capacity comparison and not as a fitted parameter-scaling law. The faint grey points show the same configurations at earlier training fractions; the colored points emphasize the full-data scores and the empirical frontier.

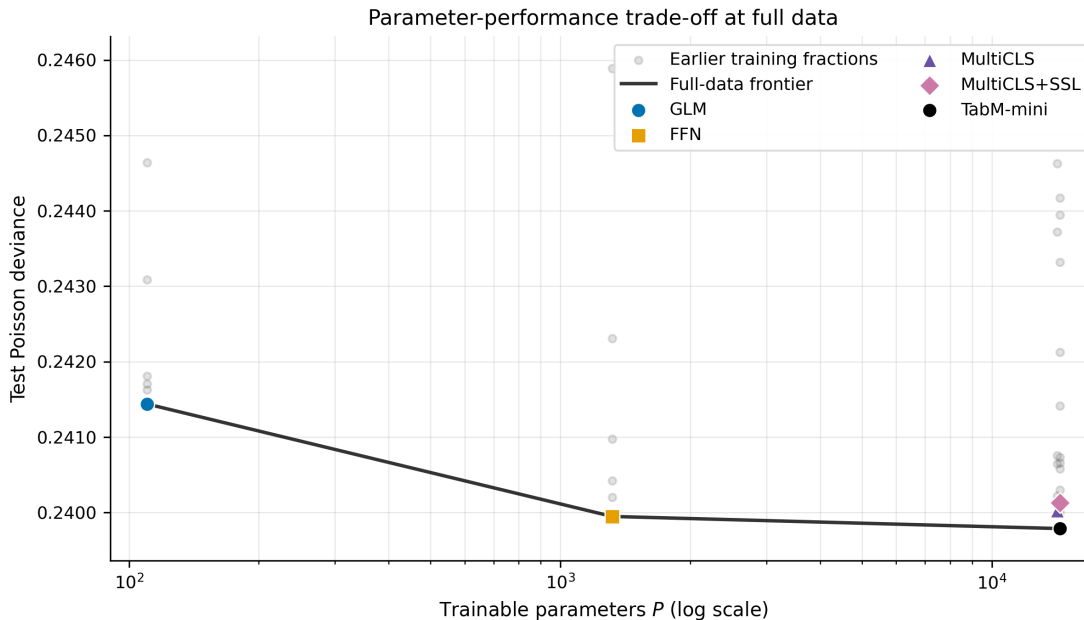


Figure 3: Parameter-performance trade-off for the FMTPL reference run. Full-data model scores are highlighted; earlier training fractions are shown as faint background points.

Figure 4 presents the same run as a likelihood lift relative to the GLM. The transformation divides the difference in Poisson deviance by two, so the vertical scale is the approximate per-policy log-likelihood lift over the GLM baseline at the same training fraction.

Figure 5 reproduces the paper’s practical-significance view. At each training fraction it selects the best model in the reproduction and reports its per-policy log-likelihood lift relative to the GLM. This view answers the actuarial question more directly than a raw loss plot: it shows how much likelihood is gained over the transparent baseline as the experiment moves from small to larger training samples.

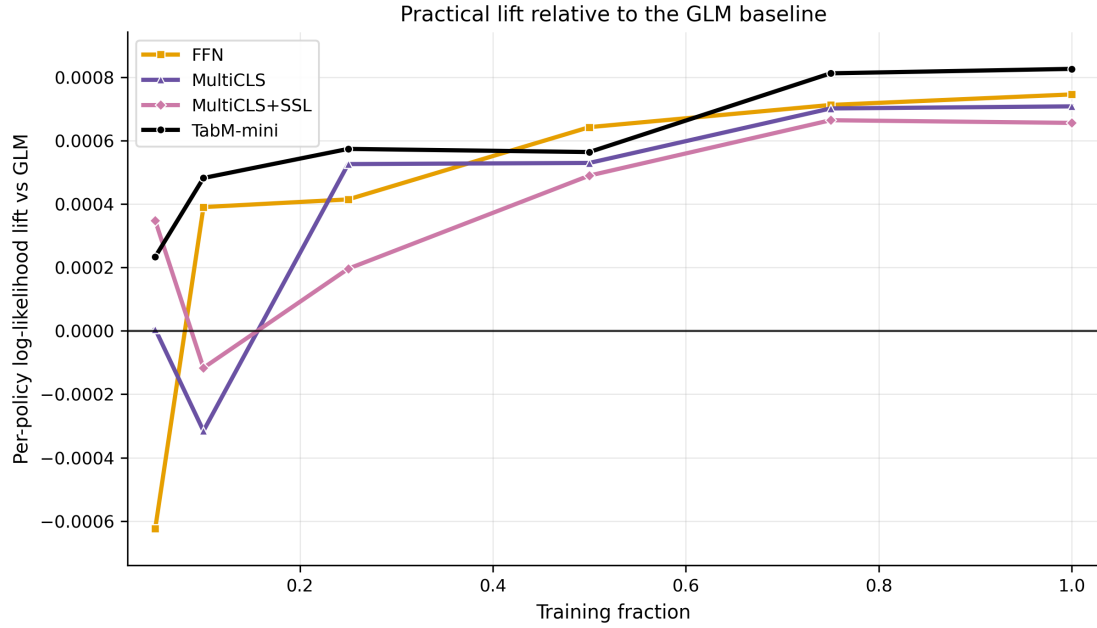


Figure 4: Approximate per-policy log-likelihood lift over the GLM baseline for the FMTPL reference run.

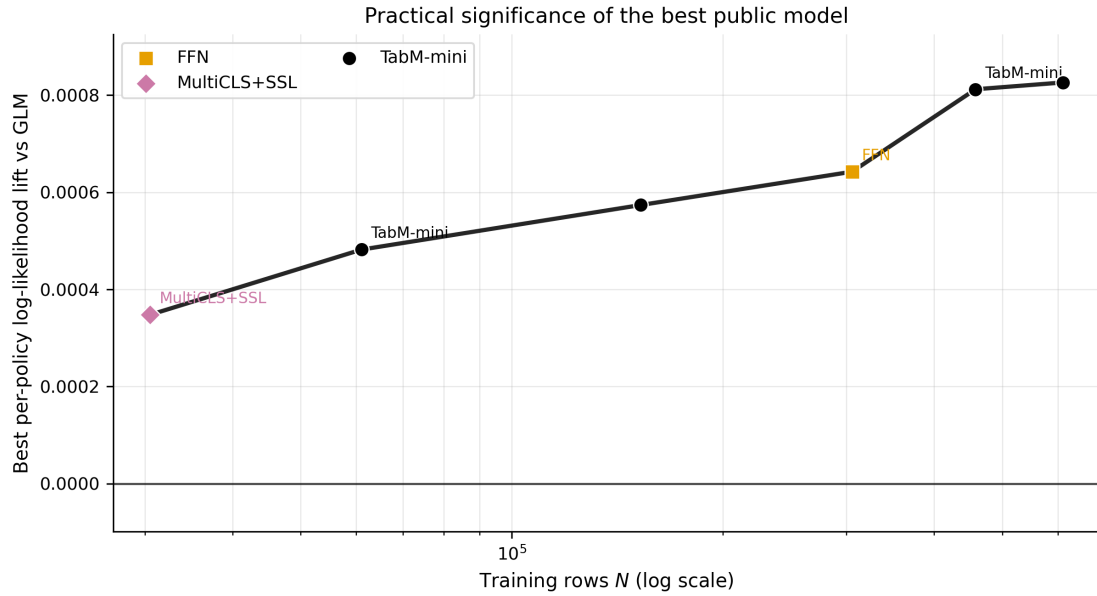


Figure 5: Best-model practical significance for the FMTPL reference run. The vertical axis is the best available per-policy log-likelihood lift over the GLM baseline at the same training fraction.

1.7 Model-Outcome Diagnostics for Business Interpretation

Likelihood gains are useful only if they correspond to model-output differences that can be interpreted by actuarial and business audiences. We therefore compare the full-data GLM and TabM-mini predictions by exposure-weighted actuarial cells. The diagnostic estimates are based on a separate full-data refit and may differ slightly from Table 2 because of stochastic training and the smaller three-seed protocol, but the qualitative conclusion is unchanged.

On the test split, the GLM predicts 2,660 claims and the diagnostic TabM-mini ensemble predicts 2,703 claims, compared with 2,645 observed claims over 35,967 exposure-years. Thus the aggregate TabM-mini minus GLM difference is about 43 claims, or 1.2 claims per 1,000 exposure-years. This is a modest portfolio-level difference. In pricing work, an overall level calibration could remove this aggregate difference if the target premium level is fixed. The more relevant business question is whether the *shape* of the indicated frequencies changes in recognizable segments.

Table 4 gives one such view. The largest age-band movement is for young drivers: the GLM predicts 15.24 claims per 100 exposure-years for ages 18-24, whereas TabM-mini predicts 16.51, much closer to the observed 16.78 in this holdout sample. For most middle-age bands, the two models are much closer. This is a useful practical interpretation of the deviance improvement: the neural model is not merely improving an abstract score; it changes selected cell-level frequency indications while leaving the broad portfolio level relatively stable.

Driver age band	Exposure years	Observed frequency claims per 100 exposure-years	GLM frequency claims per 100 exposure-years	TabM-mini frequency claims per 100 exposure-years	TabM-mini - GLM claim difference claims per 1,000 exposure-years
18-24	1,263	16.78	15.24	16.51	12.8
25-29	2,552	8.50	9.60	9.54	-0.6
30-39	8,322	7.20	7.00	7.07	0.8
40-49	8,738	6.81	7.51	7.57	0.6
50-59	8,015	7.22	6.90	6.93	0.4
60-69	4,066	6.66	6.27	6.46	1.9
70+	3,010	5.71	5.85	6.02	1.8

Table 4: Exposure-weighted frequency outcomes by driver age band on the FMTPL test split. The final column reports the TabM-mini minus GLM predicted claim difference per 1,000 exposure-years.

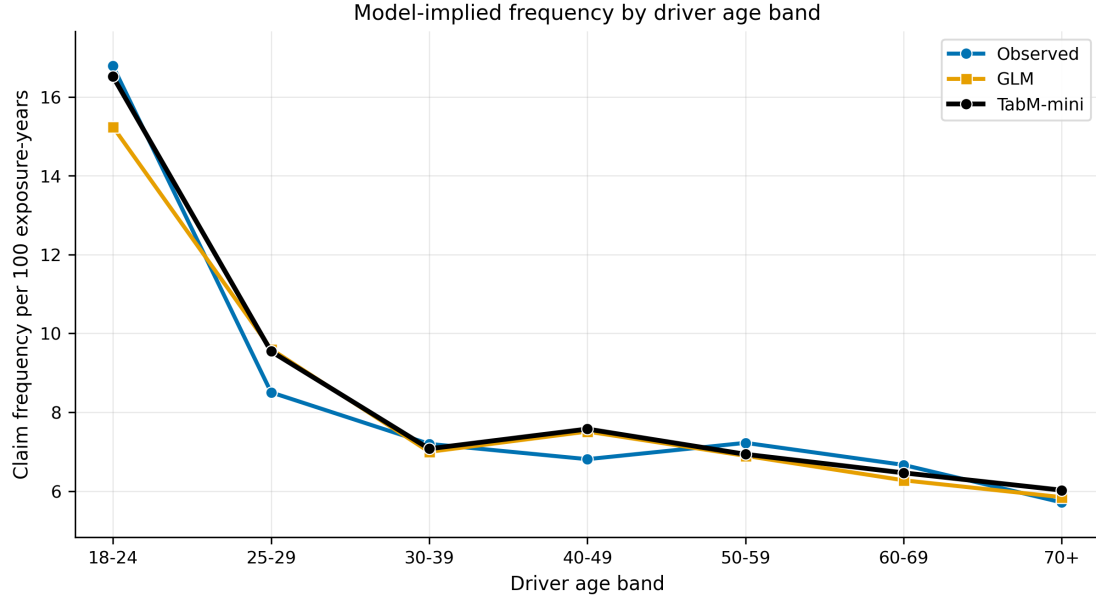


Figure 6: Observed, GLM-predicted and TabM-mini-predicted claim frequency by driver age band. Frequencies are exposure-weighted and reported as claims per 100 exposure-years.

The second diagnostic asks where TabM-mini changes the *interaction* between age and bonus-malus status. The GLM in this experiment is log-additive in the binned features, so it can learn a driver-age effect and a bonus-malus effect but not a separate age-by-bonus-malus interaction. TabM-mini can learn such interactions through its embedding MLP. Figure 7 shows the TabM-mini minus GLM predicted claim difference by age band and bonus-malus band, in claims per 1,000 exposure-years. Cells with less than 100 exposure-years are omitted.

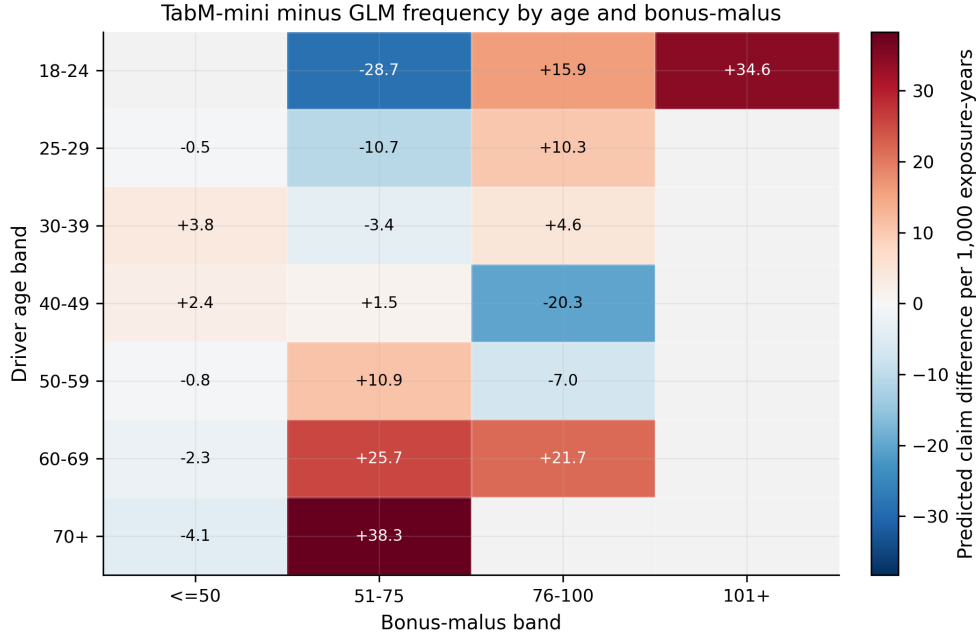


Figure 7: TabM-mini minus GLM predicted frequency by driver age band and bonus-malus band. Values are predicted claim differences per 1,000 exposure-years; blank cells have less than 100 exposure-years in the test split.

Figure 8 complements the frequency-difference heatmap by showing whether the cell-level movements improve the held-out likelihood. For instance, the 18-24 and bonus-malus 101+ cell has both a materially higher TabM-mini indicated frequency and a clear row-level deviance improvement relative to the GLM. Other cells improve because TabM-mini lowers the indicated frequency, such as ages 40-49 with bonus-malus 76-100.

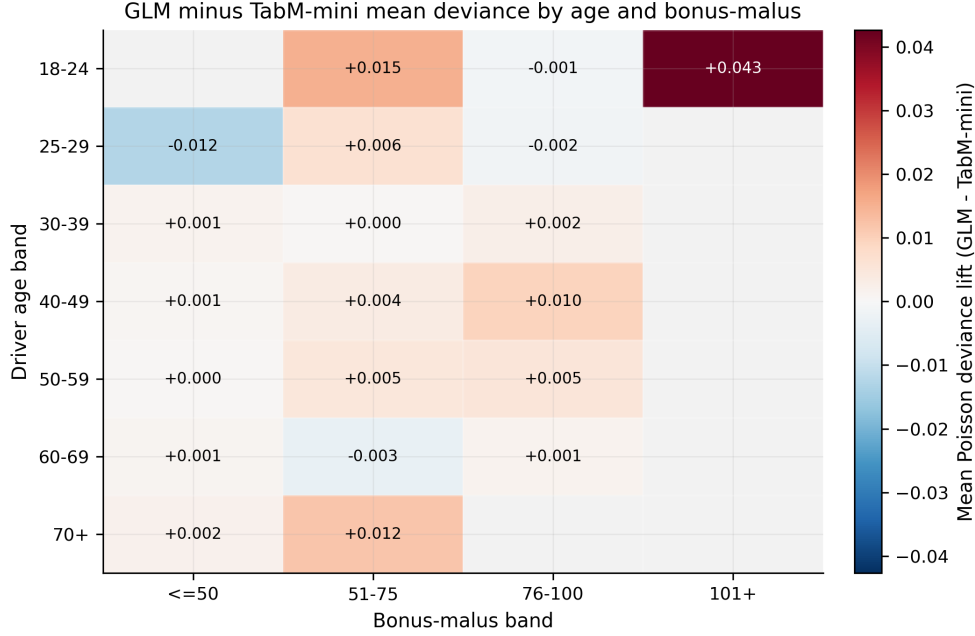


Figure 8: GLM minus TabM-mini mean Poisson deviance by driver age band and bonus-malus band. Values are row-level mean deviance lifts; positive values indicate lower mean Poisson deviance for TabM-mini. Blank cells have less than 100 exposure-years in the test split.

The interaction view gives a concrete example of how the improved deviance changes model outcomes. For example, TabM-mini is materially higher than the GLM for drivers aged 70+ with bonus-malus 51-75, and for drivers aged 18-24 with bonus-malus above 100; it is materially lower for drivers aged 18-24 with bonus-malus 51-75 and for drivers aged 40-49 with bonus-malus 76-100. These are not proposed production rating factors, and several cells are small enough that a pricing team would need credibility smoothing, regulatory review, and out-of-time validation before acting on them. They do, however, demonstrate that the deviance improvement corresponds to concrete changes in implied frequency shapes, not only to a better aggregate score.

1.8 Limitations of the FMTPL Reproduction

The FMTPL reproduction is deliberately smaller than the paper’s main experiment. It uses fewer rows, fewer covariates, smaller models, and a lower seed count. It also focuses only on frequency modeling and does not attempt to reproduce the paper’s full parameter- and compute-scaling analysis. Consequently, the run should be viewed as a transparent reproduction of the modeling and scaling-law workflow, while the empirical conclusions of the paper remain based on the larger proprietary portfolio and the broader model ladders described in the main text.

2 Using the GitHub Repository

The repository at <https://github.com/RonRichman/frmtpl-scaling-laws> is intended to let readers inspect and rerun the workflow behind this supplement. The most direct route is the Colab notebook, `notebooks/01.colab_frmtpl_scaling.ipynb`. Opening the notebook from

GitHub starts from the same code used to produce the reference run; the first setup cell clones the repository if needed, installs the local package, and checks that TensorFlow can see a GPU. If Colab reports that no GPU is available, choose `Runtime > Change runtime type > Hardware accelerator > GPU`, restart the session, and rerun the notebook.

The repository is organized around a small Python package and a few command-line scripts. `src/frmtpl_scaling/` contains the data loading, preprocessing, model definitions, training loop, loss functions, scaling-law fits, and plotting utilities. `scripts/prepare_wuthrich_data.R` rebuilds `data/FRMTPPL.csv` from the corrected `freMTPPL2freq.rda` file if the CSV is missing. `scripts/run_experiment.py` runs the nested-fraction model sweep, `scripts/make_figures.py` regenerates the scaling and likelihood-lift figures, and the diagnostics script rebuilds the GLM versus TabM-mini business diagnostics. The `results/` directory contains the reference CSV outputs, while `figures/` contains the plots reproduced in this supplement.

For a local run from a fresh checkout, first create the environment:

```
git clone https://github.com/RonRichman/frmtpl-scaling-laws.git
cd frmtpl-scaling-laws
python -m venv .venv
source .venv/bin/activate
pip install -e ".[dev]"
```

To run the full reference sweep, use:

```
python scripts/run_experiment.py
python scripts/make_figures.py
```

The full run trains five compact model families across six nested training fractions and three seeds per model/fraction. A quicker smoke-test option is documented in the repository for checking the software environment, data shape, and output-writing pipeline. To regenerate the grouped outcome diagnostics and the age-band interaction figures, run:

```
python scripts/create_outcome_diagnostics.py
```

The main outputs to inspect are `results/run_scores.csv`, which records one row per seed, model and fraction; `results/ensemble_scores.csv`, which scores the seed-averaged predictions; `results/scaling_fits.csv`, which contains the fitted data-scaling summaries; and the figures in `figures/`. The diagnostic script also writes grouped summaries under `results/outcome_diagnostics/`. These files are deliberately small enough to version-control and compare across runs. The repository also includes unit tests, so readers modifying the workflow can run `pytest` after installation to check the basic data, preprocessing, loss, model-building and command-line paths.

The knobs most useful for experimentation are the number of epochs, the seed count, the nested training fractions, the model list, and whether the GPU requirement is enforced. Changing these settings is useful for learning how the workflow behaves, but the fitted exponents and rankings should then be read as a new experiment rather than as the reference run reported above.

References

- Dutang, C., & Charpentier, A. (2026). *French motor third-party liability datasets: fremtpl*. CAS-datasets documentation. Retrieved from <https://dutangc.github.io/CASdatasets/reference/freMTPL.html> (Accessed 2026-04-30)
- Gorishniy, Y., Kotelnikov, A., & Babenko, A. (2024). *Tabm: Advancing tabular deep learning with parameter-efficient ensembling*. Retrieved from <https://arxiv.org/abs/2410.24210>
- Wüthrich, M. V., & Merz, M. (2023). *Statistical foundations of actuarial learning and its applications*. Springer Cham. Retrieved from <https://link.springer.com/book/10.1007/978-3-031-12409-9> (Open access; see Appendix B for the freMTPL2 correction and cleaning) doi: 10.1007/978-3-031-12409-9